

Credit Evaluation Ensemble Model with Self-Contained Shunt

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Abstract—In the actual loan scenario, the proportion of “good” and “bad” samples is usually extremely imbalanced. Therefore, this paper constructs an ensemble model with a pre-judging mechanism for the imbalanced datasets of small enterprise. It can learn the structure of data set independently and adjust the structure spontaneously through the “data shunt”, which can improve the efficiency of model, simplify the complexity of model and optimize the indicators to achieve the goal of reducing bank operating costs and bad debt rates.

(1) Supervised data splitter: In this paper, we select the density-based outlier test algorithm Local Outlier Factor (LOF) as the main algorithm of the shunt and the raw data will be reduced and clustered by Principal Component Analysis (PCA) and K-Means++ clustering algorithm, and the outlier test pipeline will be constructed for each cluster data. Finally, the samples identified as abnormal in each cluster will be sent to the “strict channel” to simplify the whole model, speed up the calculation and improve the recall rate and other indicators, whereas the samples identified as normal will be sent to the “loose channel” to speed up the approval of loans to normal customers.

(2) Construction of the two-channel model: It has been verified that the “strict channel” has the best performance when using the Bagging model as the main model based on eXtreme Gradient Boosting (XGBoost), and the “loose channel” has the best performance when using the Gradient Boosting Decision Tree (GBDT) single model. In general, the empirical results show that the dual-channel credit evaluation model with shunt could achieve the effect of simplifying the whole model, speeding up the calculation, and optimizing the model indicators. In particular, it is possible to control the loan pass rate under different loan policies (loose or tighten) automatically by setting multiple sets of parameters for the “shunt”.

Index Terms—credit evaluation, ensemble learning, outlier test, imbalanced dataset

I. INTRODUCTION

A perfect credit evaluation system is conducive to providing accurate customer portraits for banks, financial institutions, etc., facilitating the provision of corresponding financial services for customers, preventing credit risks and improving the efficiency of financial resource allocation. With the rapid development of big data technology and machine learning algorithms, data-driven credit evaluation methods are driving the advancement of credit evaluation technology. Both for industry and academia, how to build a more efficient and more robust credit evaluation model to achieve a successful early warning of credit risk has become a hot spot of current research.

Due to its own characteristics, the number of “good customers” is often much more than the number of “bad customers”, which leads to the instability or imprecision of credit evaluation model. Therefore, a series of methods and algorithms to deal with imbalanced data emerged. There are two solutions to imbalanced data sets: I. algorithm improvement, such as cost-sensitive model and model. II. find solutions from the data set, which USES machine learning algorithms to deal with data sets, and the common methods are under-sampling and over-sampling. Among them, the disadvantage of under-sampling is obvious – ignoring most of the information, resulting in poor prediction effect of the model. Smote algorithm is a commonly used method in recent years. Smote algorithm randomly selects a few class samples by means of random sampling, and uses the k-nearest neighbor algorithm to select the nearest k sample points and generate new sample points through the given equation. It can quickly achieve the balance between “positive” and “negative” in this way. However, its disadvantages are also obvious. Due to its randomness, smote algorithm may select

noise points for feature generation, resulting in poor quality of new samples. When generating a new sample, the boundary of “positive” and “negative” samples may be blurred because the nearest neighbor of “negative” samples is “positive” samples. In view of these defects, Yanjie Dong(2009) [2] proposed the random-smote algorithm, which changed the generated samples from the original line of two selected samples into: generated samples in the triangle region generated by two adjacent points; Han(2005) [3] et al. proposed the borderline-smote algorithm, which only selected a few samples at the classification boundary as the “raw material” for generating new samples, thus greatly improving the quality of generating new samples and solving the problem of fuzzy boundary.

Outlier test is an unsupervised classifier, by learning from the majority class data to identify abnormal points. It has been widely used in the fields of credit card anti-fraud and floor transaction, etc., and has been very effective. Existing outlier detection methods are mainly based on classification [5][6][7][8][9], nearest neighbor [10][11][12][13][14] and clustering [15][16][17][18][19][20]. Neural Network (NN) and Support Vector Machine (SVM) are the commonly used methods of outlier test based on classification, which are easy to understand and suitable for normal data sets. But the limitation is that they are all supervised learners. At present, most research of scholars are based on neighbor and clustering model. Among these methods, ESTER et al.(1996) [18] proposed a density-based spatial clustering (DBSCAN) for noise applications and achieved significant success. However, it is difficult to determine the required parameters, one is the maximum radius of the neighborhood and the other is the minimum number of data points contained in such a neighborhood, both of which are sensitive to outliers. DENG et al.(2015) [20] put forward a fast OPTICS algorithm improved from DBSCAN algorithm to generate clustering implicitly. It simply sorts these objects in a collection to get an ordered list, with the main advantage of being insensitive to input parameters. The local exception factor (LOF) algorithm proposed by BHATT et al.(2016) [13] aims to specify an outlier for each object, and it is more meaningful than the method that treats outliers as binary attributes.

Ensemble learning method is an important method to improve the prediction accuracy in data mining and machine learning. Data is the basis of all problems, and features are the raw materials of machine learning modeling, data and features determine the upper limit of machine learning. Therefore, the more complete feature processing and derivative work, the better the machine learning model is built. The thought of ensemble learning was first proposed by Dasarthy and Sheela (1979), and the ensemble learning derives from the equivalence principle of strong learning and weak learning proposed by Kearns and Valiant (1989). This method combines multiple base learners to form a strong learner, which has better generalization capabilities than a single learner because there is no doubt that multi-expert decisions are more effective than individual expert decisions. Wei Cao et al. (2018) [23] compared and analyzed the applicability of different ensemble

methods in the early warning of credit risk based on the actual data of China’s online lending platform. Xiaogang Li (2018) [24] studied the modeling of credit card data based on ensemble learning model, and finally obtained a high model accuracy. Previous studies have shown that the performance of ensemble learning model would be much better than that of a single learner if the accuracy of base learners is not too low and there must sufficient difference between base learners. Wang etc. (2018) [38] proposed an ensemble learning algorithm based on stochastic subspace, which uses random subspace method to select features of data, and the experimental results demonstrate that the performance of the ensemble algorithm based on stochastic subspace is superior to the logistic regression, decision tree, artificial neural network and SVM algorithms. Liu (2017) [39] attempts to use principal component analysis and boxplot to diagnostic outliers in the robust chain ladder method, providing a basis for insurance companies to reserve more accurate reserves.

In previous studies, the credit evaluation model based on machine learning seldom noticed that the data sets are often extremely unstable. Therefore, in the process of model training, the model would obviously pay more attention to the features related to “good customers” which occupy the majority of the whole sample. Although the model has a good ability to distinguish “good customers”, it often ignores the ability of the model to distinguish “bad customers”, which leads to the fact that the model couldn’t identify and warn the “bad customers” well in the actual application and the forecast results may bring hidden risks and increase the bad debt rate to the banks. In order to solve this problem, we consider using the KMeans++ algorithm to excavate unbalanced data, and this paper designs a two-channel credit evaluation model with a shunt to improve the model. Considering the requirements of the performance improvement of the ensemble learning method, the credit evaluation model constructed in this paper is divided into two stages. The pre-judgment is first performed using the outlier test to offload the unbalanced data set. Then enter the credit evaluation stage under the ensemble learning framework to build a more efficient and more generalized credit evaluation model.

II. CREDIT EVALUATION MODEL

A. Preparations for Model Building

1) *Logistic regression*: Logistic regression is a generalized linear model and a basic component of machine learning algorithm. It is often used for binary classification problems. Due to its high computational efficiency, easy to understand and strong interpretability, it has become one of the most popular personal credit assessment methods in the financial field. The main idea is to establish a regression formula for the classification boundary line based on the existing data, and use the optimization algorithm to find the best fitting parameter set, so as to predict and classify the data set. Given a sample set of customers (x_i, y_i) , x_i represents the feature vectors of the customer, y_i represent category labels, 0 or 1, among which $y_i = 0$ indicates that the i -th customer does not default,

$y_i = 1$ indicates that the i -th customer will default. Then the probability of the i -th customer default is

$$p(y = 1|x) = 1 - p(y = 0|x) = h(x) \quad (1)$$

$$h(x) = \frac{\exp(w^T x + w_0)}{1 + \exp(w^T x + w_0)} \quad (2)$$

where, $h(x)$ is the prediction function of logistic regression, w is the regression coefficient (weight vector) of the feature variable, w_0 is a constant term.

Then given the parameter w and the sample feature vector x , the probability that the sample belongs to each class can be uniformly written as follows

$$p(y|x, w) = h(x)^y (1 - h(x))^{1-y} \quad (3)$$

Logistic regression requires the use of maximum likelihood estimation (MLE) for parameter estimation. Since each sample is independent of each other, the likelihood function of the training sample set is

$$L(w) = \prod_{i=1}^l p(y_i|x_i, w) = \prod_{i=1}^l h(x)^{y_i} (1 - h(x))^{1-y_i} \quad (4)$$

Then its log likelihood function is

$$f(w) = \ln L(w) = \sum_{i=1}^l (y_i \ln h(x) + (1 - y_i) \ln(1 - h(x))) \quad (5)$$

Finding the maximum value of this function is equivalent to solving the following minimization problem

$$\operatorname{argmin}_w -f(w)$$

where, is also called the loss function. At this point, the gradient descent method can be used to solve the problem, and the optimization algorithm can guarantee convergence to the global optimal solution.

2) *Decision tree*: Decision tree is a common basic algorithm in machine learning. It has the advantages of easy interpretation of algorithm rules, high computational efficiency and good robustness. It is mainly used for classification and prediction. The concept of Decision Tree (CLS) was first proposed by Hunt et al. (1966). Since then, many scholars have proposed more improved algorithms, such as ID3 [25], ID4 and ID5 [26][27][28][29][30], SPRINT[31][32][33], C4.5 [34], CART [35], etc.. The decision tree is a tree structure similar to the flow chart (as shown in figure 1). It consists mainly of the root node, internal nodes, leaf nodes and branches. The data is classified by a series of rules.

The decision tree starts from the root node. First, the root node takes certain rules for comparing the values of the data samples. The greedy algorithm and the top-down method are used to judge the branches under each node (that is, each internal node can be used. Think of a test for a feature, and then divide the data sample into different sample subsets according to different results, each data sample subset constitutes a child node.

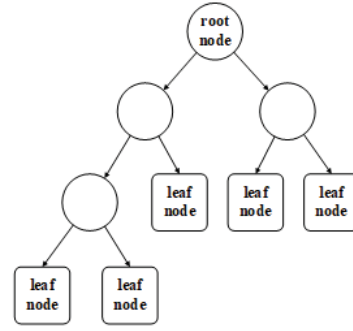


Fig. 1. Decision Tree Structure

3) *Random forest*: The random forest model is a combined classifier algorithm and it was first proposed by Leo Breiman(2001) [22]. It uses the classification regression tree (CART) as a meta-classifier, and several decision-making trees with different training sets are obtained by Bagging method. And when constructing a single tree, randomly select features to perform attribute splitting on internal nodes, and finally determine the classification result of random forest model by voting on the output of a single decision tree. The effective combination of Bagging method and CART algorithm and the existence of random disturbance make the random forest model have good generalization ability and anti-noise ability. It can not only overcome the over-fitting problem and effectively solve the unbalanced classification problem, but also shows good parallelism and scalability in the classification of high-dimensional data. The generation method of random forest is shown in figure 2 below:

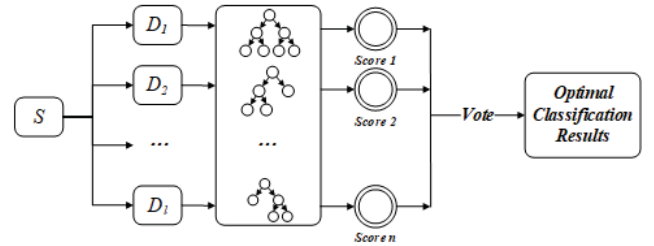


Fig. 2. Random forest algorithm

- Generate n training sets through bootstrap sampling from the original data set S ;
- Assuming that the original data set contains a total of m features, for n training sets, k feature variables in m are selected respectively, and a decision tree D_1 is generated by using CART method.
- Repeat l times and generate a total of l decision trees without pruning to form a random forest.
- Each tree in the random forest has the same distribution. Through the majority voting mechanism, all trees vote on the target to be predicted, and the classification with the most votes is the final classification.

4) *Gradient Boosting Decision Tree (GBDT)*: GBDT is an iterative Decision Tree algorithm composed of multiple decision trees, which was proposed by Jerome Friedman (1999) [36][37], and it belongs to the Boosting method of ensemble learning. The Boosting method will first assigns the same weight value to each training sample, and it would increase the weight of the classification error sample and reduce the weight of the classification correct sample when training the model in each iteration. After iterating N times, we would obtain N weak classifiers and finally integrated into a strong classifier. Different with the Boosting method, the purpose of each training of Gradient Boosting is to reduce the previous residual error. In order to continuously reduce the residual error, it is necessary to train a new model in the gradient direction of reducing the residual error. The principle of GBDT is shown in Figure 3:

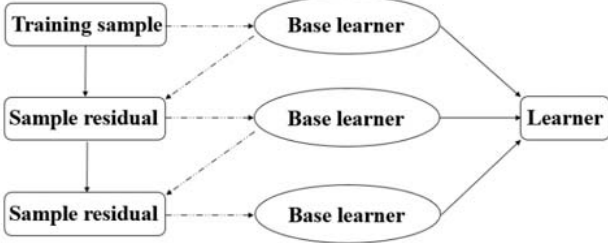


Fig. 3. GBDT algorithm

In the iteration of GBDT, assuming that the strong learner obtained in the previous iteration is $f_{t-1}(x)$ and the loss function is $L(y, f_{t-1}(x))$. The goal of our current iteration is to find a weak learner $h_t(x)$ of the CART regression tree model, so as to minimize the loss function $L(y, f_t(x)) = L(y, f_{t-1}(x) + h_t(x))$ of the epicycle, that is, finding the decision tree in the current iteration to make the loss of sample as small as possible. The core of GBDT is to let the loss function fall along the gradient direction.

5) *XGBoost (eXtreme Gradient Boosting)*: XGBoost is a distributed efficient gradient lifting algorithm based on CART decision tree. It was proposed by Chen Tianqi [1] and has attracted much attention for its superior efficiency and high prediction accuracy. This method combines multiple base learners to form a strong learner, which performs well in calculation speed and accuracy, which is significantly improved compared with GBDT and has lower computational complexity. The basic learner of this paper is based on CART regression tree (GBTree). The basic thoughts of this model are as follows:

- XGBoost algorithm combines multiple classifiers to form a strong classifier, and the final prediction result of this model is equal to the weighted sum of all CART tree prediction results.
- XGBoost algorithm adopts the same idea of GBDT that modeling based on residual error, and to get higher accuracy and reduce the probability of making mistakes

by multiple iterations.

6) *Ensemble Learning Method*: The ensemble learning method combines multiple base learners to form a strong learner, and the classic ensemble learning algorithms are Bagging and Boosting based on training data sets, and Stacking method based on base learner results. In the construction of our ensemble learning model, we will use Bagging algorithm to build our model. Bagging is an ensemble algorithm that improves classification by combining randomly generated training sets. The Bagging algorithm uses the subset of the training set as current training set to train the base learner, and after N times of training, N basic learners are formed. Then put the test set into N base learners we just have trained for prediction, and finally the N prediction results are ensemble and output. The thought of Bagging could be shown by the following figure 4:

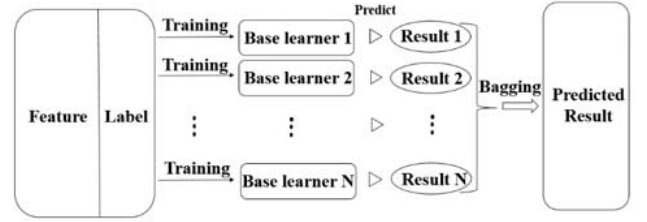


Fig. 4. Bagging algorithm

B. Construction of Credit Evaluation Model

From the above model building preparation work, we proceed to build a credit evaluation model for the unbalanced data set. The model construction is divided into the following three stages.

- The first stage (characteristic engineering): The pre-processed data set is quantified into features that the model can directly use during training and prediction through business logic understanding, data transformation, feature intersection and combination. The features are filtered from both statistics and machine learning, and finally the model features are generated.
- The second stage (pre-judgment process): KMeans clustering + outlier test is performed using the data processed by feature engineering, and a two-channel stacking model is constructed to realize the spontaneous learning and automatic adjustment of the unbalanced data set structure.
- The third stage (ensemble process): the data set after the split is trained by the same or different single model or ensemble model, and finally we integrate the prediction result of the split data set to get the final credit evaluation model.

III. EMPIRICAL ANALYSIS

A. Data Set Preprocessing

We take the loan data set of small and micro enterprises in 2016-2018 on LendingClub, the world's largest P2P platform, as the research object. First, the data set is preprocessed by

feature engineering. The base table after data preprocessing contains 8465 samples and 145 features. The ratio of “good” to “bad” samples is about 7:1.

From raw data to wide table, it has gone through three steps of imputing missing values, manual filtering and feature engineering. The specific process is as follows

- Impute missing values
Remove features with null values more than 70
- Manual filtering
Twelve features such as addr_state(region) and disbursement_method are removed, and all the remaining features are structured, and the normalization process adopts the line normalization method.
- Feature Engineering
The method of correlation analysis, feature generation and random forest was adopted for feature screening. Finally, 75 features were selected and divided into three parts: 1) Credit history; 2) Company information; 3) Cross feature of credit situation and company information.

B. Small Business Credit Evaluation Model

1) *pre-judgment process*: The outlier test model used in this paper is KMeans++ and LOF: A density-based outlier test by KMeans++ shunt, and it is used as a supervised predictor in this model to offload the data set. The KMeans++ algorithm is used to divide the data set into different clusters and enter different outlier test pipelines. After pre-judging by outlier test, the samples predicted to be “bad” in the 5 clusters are summarized as “rigorous sets”. The “rigorous” model is used for subsequent prediction classification; the samples predicted to be “good” are summarized as “loose sets”, and the “loose” models are diverted to predictive classification of subsequent integrated models.

a) *KMeans++ clustering*: Before the clustering, in order to speed up the operation, this paper uses principal component analysis (PCA) to map all samples to the two-dimensional space, and finally generates two principal components while retaining the original feature 60% information, which is used to replace all input model features, and participate in the construction of the next outlier test model. The 5 cluster centroids are as follows

TABLE I
THE 5 CLUSTER CENTROIDS

	PCA dimension 1	PCA dimension 2
Cluster 1	0.548	-0.102
Cluster 2	-0.308	-0.145
Cluster 3	0.167	-0.226
Cluster 4	0.371	0.171
Cluster 5	-0.297	0.122

b) *Construction of outlier test model*: We use LOF (density-based outlier test algorithm) to construct a supervised data shunt model for the 5-cluster sample set. Then each cluster is divided into two parts which will be sent to the loose pipe or rigorous pipe separately.

TABLE II
THE DISTRIBUTIONS OF CLUSTERS OF “LOOSE SET”

	Good Sample	Bad Sample
Cluster 1	647	359
Cluster 2	1249	359
Cluster 3	637	188
Cluster 4	2024	391
Cluster 5	637	188
Total	6032	1212

TABLE III
THE DISTRIBUTIONS OF CLUSTERS OF “RIGOROUS SET”

	Good Sample	Bad Sample
Cluster 1	176	24
Cluster 2	164	36
Cluster 3	169	31
Cluster 4	47	3
Cluster 5	173	27
Total	729	121

2) Second stage (Ensemble process):

a) *Construction of the “loose” channel*: In the loose channel, we choose Logistic Regression, Random Forest, GBDT, XGBoost as the candidate models, and learn the “loose set” respectively (the established models are simply abbreviated as LR, RF, GBDT and XGBoost). Finally, the best-performing GBDT is chosen as the model for the “loose” pipe of the data set. The effects of each model are as follows Table 4 and Figure 5. As we can see in Figure 5, the model based on GBDT has the highest quality, while other 3 models perform great. The effects of each model are as follows

TABLE IV
SUMMARY OF EVALUATION RESULTS OF “LOOSE” CHANNEL LEARNER
MODEL EVALUATION INDICATORS

	LR	RF	GBDT	XGBoost
AUC	0.755	0.878	0.914	0.893
KS	0.385	0.605	0.656	0.662
Accuracy	0.692	0.836	0.873	0.818
Precision	0.308	0.501	0.590	0.471
Recall	0.693	0.753	0.760	0.850
F1 score	0.426	0.602	0.664	0.606

b) *Construction of “rigorous” pipeline*: For the samples selected by the “shunt” to the “rigorous” pipeline, we use the bagging algorithm to construct an ensemble model based on Logistic regression, Random Forest, GBDT, and XGBoost. The weak learner, based on XGBoost, has the best effect, so it is chosen as the model of “rigorous” pipeline.

The Bagging ensemble model, choosing Logistic regression as the base learner, is recorded as Bagging Logistic (BL), the Bagging ensemble model, using random forest as the base learner is recorded as Bagging RandomForest (BR), and the Bagging ensemble model, based on GBDT, is recorded as Bagging GBDT (BG), the Bagging ensemble model based on XGBoost as the base learner is called Bagging XGBoost (BX).

The indicators of each ensemble model are shown in Table 5 and Figure 6. By comparing the indexes and ROC curve, we

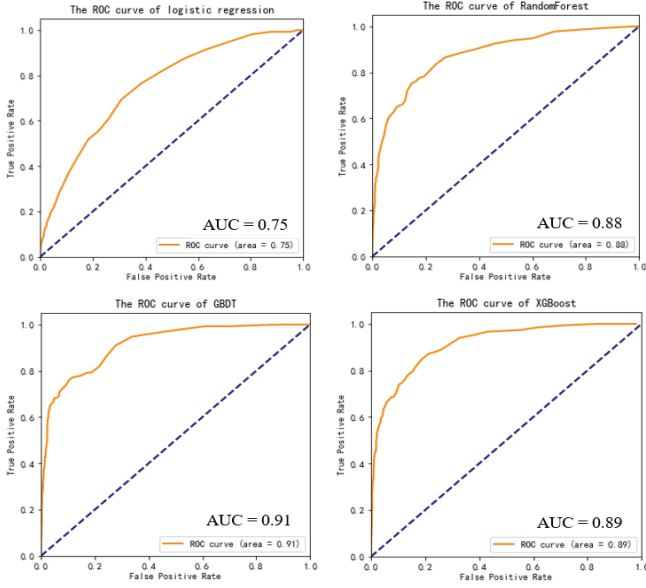


Fig. 5. ROC curve of "loose" channel learner model

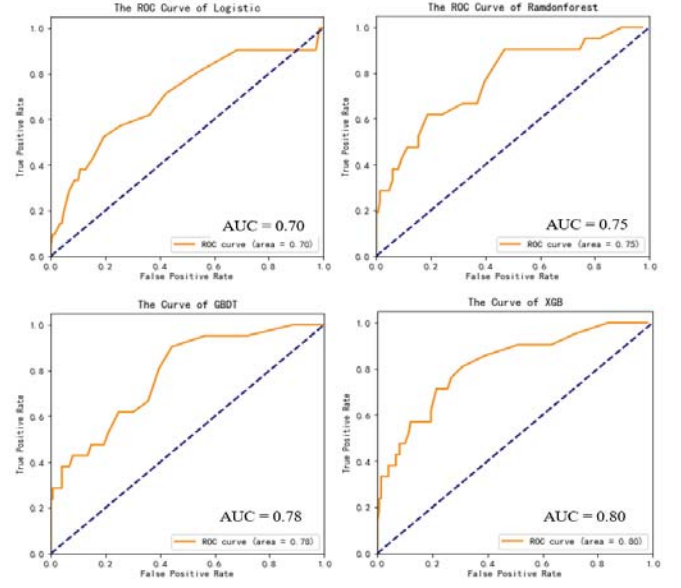


Fig. 6. ROC curve of Bagging method

can see that bagging XGBoost is the best model for "rigorous" pipeline.

The indicators of each ensemble model are as follows:

TABLE V
SUMMARY OF EVALUATION RESULTS OF EACH MODEL UNDER THE BAGGING METHOD

	BL	BR	BG	BX
AUC	0.700	0.750	0.783	0.800
KS	0.329	0.475	0.462	0.501
Accuracy	0.771	0.577	0.600	0.706
Precision	0.275	0.214	0.224	0.270
Recall	0.524	0.906	0.948	0.851
F1 score	0.361	0.905	0.359	0.405

Where, the AUC value is the area under the ROC curve and the range is $[0, 1]$. The larger the value is, the better the classification prediction effect of the model will be. The KS value is used to evaluate the model's ability to distinguish between positive and negative samples. It can be used to evaluate the model's ability to distinguish between positive and negative samples. The KS value's range is $[0, 1]$. The accuracy rate is the proportion of correct samples to the total number of samples, and the precision rate is the ratio of the number of correct positive samples to the total number of predicted positive samples. The larger their value is, the better the prediction ability of the model is. The recall rate is the proportion of correctly predicted positive samples to the real positive samples, and the larger the value, the better. F1 score is an indicator used to measure the accuracy of the binary classification model in statistics. It is the harmonic mean of model accuracy and recall rate, whose range is $[0, 1]$. The larger the value, the better.

The ROC curves for each model are as Fig.6

3) *Analysis of Result:* Through the above research, it is found that the structure and distribution of the data set have changed through the screening and shunting of the "shunt". In the "loose" data set, normal users belonging to "different groups", and the commonality of data set of each group surpasses the individuality. In the "strict" data set, most of them are "maverick" users with more personality than commonality. It is their "move-up" lifestyle that adds uncertainty to their behavior-"returning loans". Based on this, we bring together these "mavericks" and try to study their behavior and find out the commonalities. The facts show that the model's true negative rate (FPR) index has risen significantly, meanwhile the bank has maintained a stable and good level of indicators, while improving the ability to discriminate against "bad" customers. At the same time, the indicators of the "loose" model are in excellent condition.

Through empirical tests, we can see from the ROC curve of the rigorous pipeline that the curve is not smooth. This may be due to the fact that the samples flowing to this part are more difficult to distinguish. The success of distinguishing this part of the customer group is the difficulty and key to the successful establishment of the credit rating model.

In the figure 6, the abscissa axis stands for false positive rate(fpr), meanwhile the ordinate axis represents true positive rate(tpr). In the normal situation, as the fpr increases, there will be no retracement in the ROC curve. Thus choosing an appropriate threshold will be more difficult for models in such condition.

CONCLUSION

Based on the idea of establishing fast channel for "good" customers with "good behavior" and "imposing punishment" for "bad" customers with "bad behavior", we establish a two-channel credit evaluation model with shunt. Through empirical

tests, it can achieve the effect of simplifying the overall model, speeding up the calculation, and optimizing the model indicators. In particular, by setting multiple sets of parameters to the "shunt", it can automatically control the loan acceptance rate under different loan policies (loose or tight).

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